

Exactly One Digit per Height

Resolving three modular-tetration conjectures in OEIS A324017

A computer-assisted research note

Abstract

OEIS A324017 records three conjectures about the residues of $T_m = (2n - 1) \uparrow\uparrow m$ modulo $(2n)^m$. We prove the first two and disprove the third for every nontrivial base in its stated domain. The organizing result is stronger: if $q \geq 4$ is even, $T_0 = 1$, and $T_{h+1} = (q - 1)^{T_h}$, then, for every $h \geq 0$ and $r \geq 1$,

$$T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}.$$

Consequently, the tower acquires exactly one additional stable base- q digit at each height. We also determine every relevant prime-adic valuation, prove uniqueness of the modular fixed point, give a factorization-free digit-lifting algorithm, and extend the modular stabilization result to arbitrarily high Knuth arrow ranks. A local Lambert W representation explains the limiting values and the essential sign distinction at the prime 2. All main proofs are elementary and self-contained. A standard-library Python implementation and independent exact-arithmetic checks accompany the article.

Outcome and scope

Conjecture 1: true. Conjecture 2: true. Conjecture 3: false.

The smallest allowed counterexample to the third statement is

$$3 \uparrow\uparrow 2 = 27 \not\equiv 59 \equiv 3 \uparrow\uparrow 3 \pmod{64}.$$

The contribution is an explicit resolution of the statements still labelled conjectures in the retrieved OEIS entry, together with a unified sharpening. General modular stabilization and lifting-the-exponent methods are established mathematics, not claimed discoveries here.

Research status. The proofs have been checked algebraically and against independent finite computations, but have not undergone independent peer review. The literature check does not establish worldwide priority.

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1 The selected problem and its status

1.1 The source and the three precise statements

The chosen target is the set of conjectures attached to OEIS A324017, an entry introduced by Davis Smith in 2019. The entry retrieved on 19 September 2026 still labels all three statements as conjectures [1]. This is a concrete, checkable source of the problem; it is not a claim that no proof exists anywhere else.

Write

$$q = 2n, \quad a = q - 1, \quad T_0 = 1, \quad T_{h+1} = a^{T_h}. \quad (1.1)$$

Thus $T_h = a \uparrow\uparrow h$, with the exponentiation evaluated from the top of the tower downward. The array in the source is

$$A(m, n) = T_m \bmod q^m,$$

where a residue always means the least nonnegative representative. In this notation the three source conjectures are as follows.

C1: fixed point.

For $m, n, k > 1$, if $T_m \equiv k \pmod{q^m}$, then $a^k \equiv k \pmod{q^m}$.

C2: second digit.

For $n > 1$ and $m \geq 2$, $\lfloor T_m/q \rfloor \equiv q - 2 \pmod{q}$.

C3: early stabilization.

For $m > 1$, if $T_m \equiv j \pmod{q^{m+1}}$, then $A(m+1, n) = j$, with j understood as a canonical residue. Equivalently, $T_{m+1} \equiv T_m \pmod{q^{m+1}}$.

The canonical-residue convention removes an irrelevant ambiguity in C3: literal equality with an arbitrary congruent integer would already be meaningless. Even under its natural, most favorable interpretation, C3 is false. Our resolution concerns the statements as displayed; we do not speculate about a different statement that the author might have intended.

1.2 Why this is an approachable research target

The modulus is not arbitrary: the tower base is one less than an even radix. Every tower is odd, and

$$a \equiv -1 \pmod{q}.$$

These two observations neutralize the alternating sign in exponentiation. Differences of tower heights can then be studied without constructing any tower. In particular, the difference of two exponentials factors into a unit times an expression of the form $a^d - 1$, with d even. That is exactly the setting in which elementary prime-adic valuations are effective.

General stabilization of modular power towers and its algorithmic implications are part of an existing literature. Hittmeir studies the relationship between general modular tetration and integer factorization [3]. Ripà gives congruence-speed formulas for decimal integer tetration [2]. Those works are important context: proving that some digits eventually stabilize would not by itself be a new contribution. Here the target is the exact diagonal family $q = a + 1$, the exact height threshold, and the particular conjectures in [1].

1.3 A counterexample before the general theory

Take $n = m = 2$, so $q = 4$ and $a = 3$. Then

$$T_2 = 27, \quad T_3 = 3^{27} = 7\,625\,597\,484\,987.$$

Reducing modulo $q^{m+1} = 64$ gives 27 and 59, respectively. For a small independent certificate, note that

$$3^8 \equiv 33, \quad 3^{16} \equiv 1, \quad 3^{27} \equiv 1 \cdot 33 \cdot 9 \cdot 3 \equiv 59 \pmod{64}.$$

Thus C3 fails at the first pair $m, n > 1$ permitted by its nontrivial interpretation. This is not a rare anomaly: Theorem 5.2 will show that it fails for every $n > 1$ and every $m > 1$.

2 Notation and the meaning of an exact stable digit

For a prime p and a nonzero integer x , write $v_p(x)$ for the exponent of p in x . When q is composite, it is useful to introduce a separate precision order

$$\text{ord}_q(x) = \max\{s \geq 0 : q^s \mid x\} = \min_{p|q} \left\lfloor \frac{v_p(x)}{v_p(q)} \right\rfloor. \quad (2.1)$$

This is not, in general, a multiplicative valuation. For example, $\text{ord}_6(2) = \text{ord}_6(3) = 0$ but $\text{ord}_6(6) = 1$. We therefore use the phrase *base- q precision order*, reserving “valuation” for v_p .

Two nonnegative integers have the same last s base- q digits exactly when they are congruent modulo q^s . If their precision order of difference is s , then the digit in position s differs, with the units digit in position 0.

The central claim can therefore be expressed without any numerical approximation: every later tower agrees with T_h in exactly its last h base- q digits, and disagrees in the next digit. The fact that q and the tower base are different is essential. The phrase “one digit per height” always refers to radix q , not to radix a .

Throughout the main argument, $q \geq 4$ is even, so $a = q - 1 \geq 3$. The excluded case $q = 2$ has $a = 1$ and $T_h = 1$ for every height; it is trivial but does not satisfy any nonzero-difference formula.

3 Elementary valuation tools

We give the special lifting-the-exponent facts needed below, including their proofs. No theorem about real or complex tetration is involved.

Lemma 3.1. *Let p be odd and let $X > 1$ be an integer with $X \equiv 1 \pmod{p}$. For every positive integer d ,*

$$v_p(X^d - 1) = v_p(X - 1) + v_p(d).$$

The same formula holds for $p = 2$ under the stronger assumption $X \equiv 1 \pmod{4}$.

Proof. First suppose p is odd and $s = v_p(X - 1) \geq 1$. If d is coprime to p , the factor $1 + X + \cdots + X^{d-1}$ is congruent to d modulo p , so it is a unit. Thus raising X to a p -coprime exponent leaves the valuation of $X - 1$ unchanged.

Writing $X = 1 + p^s u$, with $p \nmid u$, the binomial expansion of $X^p - 1$ has first term $p^{s+1}u$. Every other term has valuation at least $s + 2$. For the last term this uses $ps \geq s + 2$, which

holds for $p \geq 3$ and $s \geq 1$. Consequently, raising to the p th power adds exactly one to the valuation. Factor $d = p^t w$, with $p \nmid w$, and apply these two observations successively.

For $p = 2$, odd exponents are handled by the same geometric sum. If $X \equiv 1 \pmod{4}$, then $v_2(X + 1) = 1$, whence $v_2(X^2 - 1) = v_2(X - 1) + 1$. Repeated squaring and the odd part of d complete the proof. $\square$

Set

$$u = v_2(q - 2) = v_2(a - 1), \quad v = v_2(q) = v_2(a + 1), \quad c = u + v - 1. \quad (3.1)$$

Both u and v are positive, one of them is 1, and $c \geq v$.

Lemma 3.2. *For every positive even integer d and every odd prime $p \mid q$,*

$$v_p(a^d - 1) = v_p(q) + v_p(d), \quad (3.2)$$

$$v_2(a^d - 1) = c + v_2(d). \quad (3.3)$$

If t is a positive odd integer, then $v_2(a^t - 1) = u$.

Proof. Write $a^d = (a^2)^{d/2}$. For odd $p \mid q$, the identity $a^2 - 1 = (q - 2)q$ gives $v_p(a^2 - 1) = v_p(q)$, because $p \nmid q - 2$. Apply Lemma 3.1; division of d by 2 does not affect its p -adic valuation.

At $p = 2$, $a^2 \equiv 1 \pmod{8}$ and $v_2(a^2 - 1) = u + v$. The same lemma gives $u + v + v_2(d/2) = c + v_2(d)$. For odd t , the geometric sum in $a^t - 1 = (a - 1)(1 + a + \cdots + a^{t-1})$ is odd. $\square$

4 Exact distance between any two tower heights

Theorem 4.1 (Exact local distances). *For every even $q \geq 4$, every $h \geq 0$, every $r \geq 1$, and every odd prime $p \mid q$,*

$$v_p(T_{h+r} - T_h) = h v_p(q), \quad (4.1)$$

$$v_2(T_{h+r} - T_h) = u + h(u + v - 1). \quad (4.2)$$

In particular, the answer is independent of the positive height gap r .

Proof. Fix r and put $D_h = T_{h+r} - T_h$. Since $a \geq 3$, the towers strictly increase, so all D_h are positive. Every tower is odd, making every D_h even.

At height 0, $D_0 = T_r - 1 = a^{T_{r-1}} - 1$. For an odd prime $p \mid q$, the oddness of T_{r-1} gives $T_r \equiv -1 \pmod{p}$, and hence $D_0 \equiv -2 \pmod{p}$. Thus $v_p(D_0) = 0$. The last assertion of Lemma 3.2 gives $v_2(D_0) = u$.

For the induction step, factor

$$D_{h+1} = a^{T_h} (a^{D_h} - 1). \quad (4.3)$$

The first factor is a unit at every prime dividing q . Since D_h is even, Lemma 3.2 adds $v_p(q)$ at each odd prime and adds c at 2. Starting from the height-zero values gives exactly (4.1) and (4.2). $\square$

For clarity, the 2-adic formula has the following two forms:

$$v_2(T_{h+r} - T_h) = \begin{cases} 1 + h v_2(q), & q \equiv 0 \pmod{4}, \\ (h + 1)v_2(q - 2), & q \equiv 2 \pmod{4}. \end{cases} \quad (4.4)$$

Corollary 4.2 (Exactly one radix digit per height). *For every $h \geq 0$ and $r \geq 1$,*

$$\text{ord}_q(T_{h+r} - T_h) = h. \quad (4.5)$$

Moreover, $2q^h \mid T_{h+r} - T_h$.

Proof. The local formulas imply $q^h \mid D_h$. At 2 they give the stronger inequality $v_2(D_h) = u + hc \geq 1 + hv$, proving $2q^h \mid D_h$.

If q has an odd prime factor p , then $v_p(D_h) = h v_p(q) < (h+1)v_p(q)$, so $q^{h+1} \nmid D_h$. If q has no odd prime factor, write $q = 2^v$ with $v \geq 2$. Then $u = 1$ and $v_2(D_h) = 1 + hv < (h+1)v$. This proves (4.5) in all cases. $\square$

Corollary 4.3 (Sharp stabilization threshold). *Fix $m \geq 1$. The residues $T_h \bmod q^m$ become constant precisely at height m . More strongly, no height $h < m$ has the eventual residue.*

Proof. If $h \geq m$, (4.5) shows that every T_{h+r} agrees with T_h modulo q^m . If $h < m$, then $T_m - T_h$ has precision order $h < m$, so T_h differs from the eventual residue $T_m \bmod q^m$. $\square$

This is stronger than saying that neighboring residues eventually agree: it rules out an earlier accidental visit to the final residue followed by a departure. The exact formulas also explain why the excess powers of 2 do not speed up the full radix- q stabilization. An odd prime factor is a bottleneck unless q is a power of 2; in that exceptional case the single extra factor of 2 is still less than an entire new radix digit.

5 The universal first unstable-digit difference

The preceding result determines the first location of disagreement. We now determine the difference at that location. A factor of 2 in Corollary 4.2 is crucial when the radix is even.

Lemma 5.1 (A binomial precision estimate). *Let $q \geq 4$ be even, let $h \geq 0$, and let $d > 0$ be divisible by $2q^h$. Then*

$$(q-1)^d \equiv 1 - qd \pmod{q^{h+2}}. \quad (5.1)$$

Proof. The exponent d is even, so $(q-1)^d = (1-q)^d$. We must show that every binomial term $\binom{d}{j}(-q)^j$ with $j \geq 2$ is divisible by q^{h+2} . For a prime $p \mid q$, put $e = v_p(q)$. The identity

$$\binom{d}{j} = \frac{d}{j} \binom{d-1}{j-1}$$

shows that the valuation of this term is at least

$$je + v_p(d) - v_p(j).$$

For odd p , this is at least $(h+j)e - v_p(j)$, which is at least $(h+2)e$ because $v_p(j) \leq j-2$ for $j \geq 2$. For $p = 2$, the extra factor 2 in d gives the lower bound $(h+j)e + 1 - v_2(j)$. It is at least $(h+2)e$, since $v_2(j) \leq j-1$ and $e \geq 1$. This holds at every prime dividing q , proving the claimed divisibility. $\square$

Theorem 5.2 (Universal leading difference). *For every even $q \geq 4$, every $h \geq 0$, and every $r \geq 1$,*

$$T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}. \quad (5.2)$$

Equivalently,

$$\frac{T_{h+r} - T_h}{q^h} \equiv q - 2 \pmod{q}. \quad (5.3)$$

Proof. Again fix r and put $D_h = T_{h+r} - T_h$. At height zero, $D_0 \equiv -2 \pmod{q}$, because every positive-height tower is congruent to -1 modulo q .

By Corollary 4.2, $2q^h \mid D_h$. Apply Lemma 5.1 to (4.3):

$$D_{h+1} \equiv a^{T_h}(-qD_h) \pmod{q^{h+2}}.$$

Since T_h is odd, $a^{T_h} \equiv -1 \pmod{q}$. Also, qD_h is divisible by q^{h+1} . Replacing the first factor by -1 is therefore legitimate modulo q^{h+2} , and gives

$$D_{h+1} \equiv qD_h \pmod{q^{h+2}}.$$

Induction from $D_0 \equiv -2 \pmod{q}$ proves (5.2). $\square$

Corollary 5.3 (Complete refutation and correction of C3). *For every $n > 1$ and $m > 1$, C3 is false. Its correct replacement is*

$$A(m+1, n) \equiv T_m - 2(2n)^m \pmod{(2n)^{m+1}}. \quad (5.4)$$

For $n = 1$, the tower is constantly 1 and the original C3 is trivially true.

Proof. Use (5.2) with $h = m$ and $r = 1$. The residue $-2q^m$ is nonzero modulo q^{m+1} because $q \geq 4$ does not divide 2. The corrected congruence is the same identity rearranged. $\square$

Remark 5.4. The number $q - 2$ is the *difference* of the first unstable digits, modulo q . It is not a claim that the digit itself always equals $q - 2$. Digits farther to the left can also change, and carries must not be ignored.

6 Modular fixed points and the two true conjectures

6.1 Why exponent reduction is valid here

Blindly replacing a tower exponent by its residue modulo the output modulus is invalid in general. For example, $2^3 \not\equiv 2^{3+3} \pmod{3}$. Our restricted family admits precisely the compatibility that such a computation needs.

Lemma 6.1 (Exponent period and contraction). *For every $s \geq 1$,*

$$a^{q^s} \equiv 1 \pmod{q^{s+1}}. \quad (6.1)$$

If x, y are nonnegative odd integers and $q^s \mid x - y$, where now $s \geq 0$, then

$$a^x \equiv a^y \pmod{q^{s+1}}. \quad (6.2)$$

Proof. For (6.1), the exponent q^s is even. The valuation formulas in Lemma 3.2 give $(s+1)v_p(q)$ at each odd prime and at least $(s+1)v_2(q)$ at 2.

For the second statement, interchange x, y if necessary and factor $a^x - a^y = a^y(a^{x-y} - 1)$. If $x = y$ there is nothing to prove. Otherwise $x - y$ is even. At an odd prime, exponentiation adds $v_p(q)$ to the valuation of the difference; at 2 it adds $c \geq v_2(q)$. The case $s = 0$ is included in this argument. $\square$

The period identity shows that

$$F_m : \mathbb{Z}/q^m\mathbb{Z} \longrightarrow \mathbb{Z}/q^m\mathbb{Z}, \quad x \longmapsto a^x$$

is well defined, when evaluated on nonnegative representatives. It always returns an odd residue. The contraction identity says that, on odd residues, one application gains at least one radix digit of agreement.

Theorem 6.2 (Unique modular fixed point). *For each $m \geq 1$, the congruence*

$$a^x \equiv x \pmod{q^m} \tag{6.3}$$

has exactly one residue-class solution. It is

$$R_m := T_m \bmod q^m.$$

The representatives satisfy

$$R_{m+1} \equiv R_m \pmod{q^m}. \tag{6.4}$$

Proof. By the period property, $a^{R_m} \equiv a^{T_m} = T_{m+1} \pmod{q^m}$. By Corollary 4.3, $T_{m+1} \equiv T_m \equiv R_m \pmod{q^m}$. This proves existence.

If x and y are two fixed points, both are odd. Starting with their agreement modulo q^0 , apply (6.2) and the fixed-point equations to obtain their agreement modulo q , then modulo q^2 , and so on through q^m . This proves uniqueness. Compatibility follows directly from the stabilization of the towers. $\square$

Corollary 6.3 (C1 is true). *For every $m \geq 1$, every $n > 1$, and every nonnegative integer k with $T_m \equiv k \pmod{(2n)^m}$,*

$$(2n - 1)^k \equiv k \pmod{(2n)^m}.$$

Thus the source's restrictions $m, k > 1$ are unnecessary.

Proof. Such a k represents the unique fixed-point class R_m . The exponent period ensures that changing its representative does not change the value of a^k modulo q^m . $\square$

Theorem 6.4 (Explicit two-digit residue; C2 is true). *For $n > 1$ and every $h \geq 2$,*

$$T_h \equiv q^2 - q - 1 \pmod{q^2}. \tag{6.5}$$

In particular, $\lfloor T_h/q \rfloor \equiv q - 2 \pmod{q}$.

Proof. The exponent $a = q - 1$ is odd. A first-order binomial expansion gives

$$T_2 = (q - 1)^{q-1} = -(1 - q)^{q-1} \equiv -1 + q(q - 1) \equiv -1 - q \pmod{q^2}.$$

Its canonical representative is $q^2 - q - 1$. All heights $h \geq 2$ have the same residue modulo q^2 by Corollary 4.3. Finally,

$$q^2 - q - 1 = q(q - 2) + (q - 1).$$

Dividing T_h by q and reducing the integer quotient modulo q therefore gives $q - 2$. $\square$

7 The limiting integer and its digits

7.1 An inverse limit, not a real infinite tower

The compatible residues R_m define an element

$$\xi_q \in \mathbb{Z}_q := \varprojlim_m \mathbb{Z}/q^m \mathbb{Z}.$$

For composite q , this is a complete ring rather than a field. By the Chinese remainder theorem, it can be identified with $\prod_{p|q} \mathbb{Z}_p$. We do not interpret ξ_q as a finite real limit: since $a \geq 3$, the ordinary real tower diverges to infinity. Only its modular tails converge.

Proposition 7.1. *For every $h \geq 0$,*

$$\xi_q - T_h \equiv -2q^h \pmod{q^{h+1}}, \quad \text{ord}_q(\xi_q - T_h) = h. \quad (7.1)$$

Each local component $\xi_{q,p}$ also satisfies the exact valuations in Theorem 4.1, with T_{h+r} replaced by $\xi_{q,p}$.

Proof. Every sufficiently late tower represents ξ_q at any fixed precision. Thus the congruence in Theorem 5.2 passes directly to the inverse limit. For a prime $p \mid q$, use precision one higher than the valuation in Theorem 4.1. All later differences are nonzero modulo that prime power, and convergence makes their residue eventually constant; its valuation is therefore unchanged. $\square$

7.2 Digit lifting without building towers

The next theorem gives a simple recurrence for the array itself.

Theorem 7.2 (Canonical digit lift). *The fixed-point representatives are obtained by*

$$R_1 = q - 1, \quad R_{m+1} = a^{R_m} \pmod{q^{m+1}}. \quad (7.2)$$

Equivalently, if $R_{m+1} = R_m + d_m q^m$ with $0 \leq d_m < q$, then

$$d_m \equiv \frac{a^{R_m} - R_m}{q^m} \pmod{q}. \quad (7.3)$$

The numerator in the quotient is divisible by q^m .

Proof. Since $T_m \equiv R_m \pmod{q^m}$ and both are odd, Lemma 6.1 gives $a^{T_m} \equiv a^{R_m} \pmod{q^{m+1}}$. The left side is T_{m+1} , proving (7.2). Compatibility gives (7.3); the fixed-point property at precision m proves the divisibility of its numerator. $\square$

The repeated value $R_3 = R_4 = 59$ at $q = 4$ is not a counterexample to exact one-digit stabilization. It says that one digit of the limit is zero. It does *not* say that T_3 is already correct modulo 4^4 : in fact $T_3 \equiv 187 \pmod{256}$, whereas the stable residue is 59. Confusing an already truncated representative R_m with the full tower T_m would conceal exactly the discrepancy that refutes C3.

Table 1: Computed stable representatives R_m . The modulus in each column is q^m , so all entries are ordinary decimal displays of different-radix residues.

m	$q = 4$, base 3	$q = 6$, base 5	$q = 10$, base 9
1	3	5	9
2	11	29	89
3	59	29	289
4	59	1109	5289
5	827	3701	45289
6	2875	34805	745289
7	15163	128117	2745289
8	31547	687989	92745289

8 Algorithms and independent exact verification

8.1 A factorization-free evaluator

The following is the essential stable-residue algorithm, reproduced in the accompanying code/tower_digits.py with input validation.

```
def stable_residue(q, precision):
    a, modulus, x = q - 1, q, q - 1
    for _ in range(2, precision + 1):
        modulus *= q
        x = pow(a, x, modulus)
    return x
```

To evaluate $T_h \bmod q^m$ for a finite height, first replace h by $\min(h, m)$. At a fixed modulus $M = q^m$, iterate $x \leftarrow a^x \bmod M$ from $x = 1$ that many times. The period in Lemma 6.1 justifies reducing each exponent modulo M ; Corollary 4.3 justifies the height cap. The implementation uses digit lifting for the stabilized case and the fixed-modulus loop for smaller heights.

Let $L = \lfloor \log_2(q^m) \rfloor + 1$ be the output-modulus bit length. There are at most m modular exponentiations, and every exponent stored has at most L bits. With ordinary schoolbook multiplication and reduction, square-and-multiply gives the safe bit-complexity bound

$$O(mL^3)$$

and $O(L)$ working storage for a single evaluation, apart from optional caching of completed queries. This is a deliberately conservative bound, not a claim of optimality. Crucially, the computation does not factor q , construct T_h , or depend on a tower-sized exponent. The restriction $a = q - 1$ is why this does not provide an unrestricted fast modular-tetration or factoring algorithm.

8.2 An independent oracle

A test based solely on the same fixed-modulus iteration could reproduce the same mistake as the implementation. The verification script therefore uses a second method: a totient chain with exact capped exponents. For a general modulus M , put $P = \varphi(M)$. If the true exponent E is less than P , it is computed exactly. Otherwise it is replaced by

$$(E \bmod P) + P.$$

Why is this safe even for a nonunit base? At a prime power dividing M where the base is a unit, the relevant multiplicative period divides P . At a prime power p^e where the base is divisible by p , both exponents are at least $P \geq \varphi(p^e) \geq e$, so both powers vanish modulo p^e . The Chinese remainder theorem finishes the justification.

The oracle is itself checked against directly constructed, manageable integer towers at thousands of small parameter values. Thus the validation has three levels: literal integers, a general modular oracle, and the specialized algorithm.

8.3 Recorded checks

The supplied run completed **62,649 exact assertions**. It covered every even q from 4 through 200, heights 0 through 10, gaps 1, 2, 3, and output precisions through 10, in the combinations detailed in `data/verification.json`. It also examined all **51,414 residue candidates** in a separate exhaustive fixed-point search for even $q \leq 24$ and precisions 1, 2, 3.

Table 2: Selected assertion counts from the reproducible test report.

Check family	Assertions
General oracle against literal integers	8700
Universal leading-difference congruence	3267
Exact prime-adic distances	7260
Specialized tower residues against the oracle	10890
Fixed-point representatives	2970
C3 fails throughout the tested nontrivial range	891
Capped hyperoperations against small exact values	21000
Local Lambert-series residues	60

All these checks use integer or rational arithmetic; no floating-point agreement is being mistaken for proof. The finite computations are supporting evidence and software validation. The infinite statements follow from the preceding proofs, not from extrapolation from a search.

9 From tetration to higher Knuth arrows

The selected problem lies directly in tetration, but the same result also makes certain higher hyperoperations tractable modulo q^m . We state all normalization conventions to avoid an arrow-count shift.

For $a = q - 1 \geq 3$, define

$$\begin{aligned}
 H_1(n) &= a^n, \\
 H_r(0) &= 1 & (r \geq 2), \\
 H_r(n+1) &= H_{r-1}(H_r(n)) & (r \geq 2).
 \end{aligned}$$

Thus $H_r(n) = a \uparrow^r n$ and $H_2(n) = T_n$.

Lemma 9.1 (Every higher operation is an integer-height tower). *There are integers $J_r(n) \geq 0$, for $r \geq 2$, such that*

$$H_r(n) = T_{J_r(n)}.$$

They can be specified by

$$J_2(n) = n, \quad (9.1)$$

$$J_r(0) = 0, \quad J_r(n+1) = J_{r-1}(H_r(n)) \quad (r \geq 3). \quad (9.2)$$

Furthermore, $J_r(1) = 1$ and

$$J_r(n) \geq n + r - 2 \quad (r \geq 2, n \geq 2). \quad (9.3)$$

Proof. For $r = 2$ the representation is the definition of tetration. Assuming it at rank $r - 1$,

$$H_r(n+1) = H_{r-1}(H_r(n)) = T_{J_{r-1}(H_r(n))},$$

which proves the representation at rank r . The identities at $n = 0, 1$ follow immediately.

An induction on rank and argument gives $H_r(n) \geq a^n$ for every $r \geq 1$ and $n \geq 0$. Indeed, at rank r the recursion yields $H_r(n+1) \geq a^{H_r(n)} \geq a^{n+1}$. For $n \geq 2$ and $a \geq 3$, one has $a^{n-1} \geq n+1$. Now induct on r in (9.3). The case $r = 2$ is equality. At rank $r \geq 3$, the argument $H_r(n-1)$ is at least $n+1 \geq 3$, so

$$J_r(n) = J_{r-1}(H_r(n-1)) \geq H_r(n-1) + r - 3 \geq n + r - 2.$$

□

Theorem 9.2 (Modular saturation across arrow ranks). *For $r \geq 2$ and $n \geq 0$,*

$$H_r(n) \equiv R_m \pmod{q^m} \iff J_r(n) \geq m. \quad (9.4)$$

In particular, for $n \geq 2$ the readily checkable condition

$$n + r - 2 \geq m \quad (9.5)$$

is sufficient. Thus, at fixed precision, sufficiently high arrow ranks give exactly the same modular tail as a sufficiently tall tower.

Proof. The equivalence follows from the representation in Lemma 9.1 and the sharp threshold in Corollary 4.3. The sufficient condition follows from (9.3). □

For example, the supplied implementation evaluates $9 \uparrow^{1000000} 2$ modulo 10^{60} as

$$\begin{aligned} &864868894914047889985007525482 \\ &726503475610748087597392745289, \end{aligned}$$

where the two lines concatenate to the 60-digit decimal residue. This result is simply R_{60} for $q = 10$; the million-arrow expression is never expanded.

For queries below the sufficient saturation bound, the code uses exact capped hyperoperations. To compute $\min(H_r(n), C)$, let t be the least integer with $a^t \geq C$. The inequalities

$$H_r(n) \geq a^n, \quad H_r(n) \geq a^r \quad (r \geq 2, n \geq 2)$$

allow immediate return of C whenever $n \geq t$ or $r \geq t$, after handling $n = 0, 1$. Otherwise the small recursion can be evaluated without creating any integer exceeding the cap, apart from bounded intermediate products used to determine the threshold. In a modular query the needed cap is only m .

The assumption $a \geq 3$ matters. One must not extend the rank bound to $a = 2$: $2 \uparrow^r 2 = 4$ for every positive rank r . Nor does Theorem 9.2 assert that arbitrary Conway chains or arbitrary fast-growing-hierarchy notations have been evaluated. It is a precise extension to Knuth hyperoperations with the stated base and modulus relationship.

10 A local Lambert W description

This optional section expresses the limiting residue system using p -adic analysis. It is not needed for any of the three conjectures. The p -adic Lambert function and its power series are established objects; see Mező [4]. Our task is to select the correct local sign for this tower family.

10.1 Principal units and the sign at two

Fix a prime $p \mid q$ and define

$$\varepsilon_p = \begin{cases} +1, & p = 2 \text{ and } q \equiv 2 \pmod{4}, \\ -1, & \text{otherwise,} \end{cases} \quad b_p = \varepsilon_p(q-1), \quad \ell_p = \log_p(b_p). \quad (10.1)$$

For odd p , $b_p = 1 - q \in 1 + p\mathbb{Z}_p$. For $p = 2$, the definition ensures $b_p \in 1 + 4\mathbb{Z}_2$. Consequently logarithm and exponential are inverse on the relevant principal-unit neighborhoods. More explicitly,

$$v_p(\ell_p) = \begin{cases} v_p(q), & p \text{ odd,} \\ v_2(q), & p = 2, q \equiv 0 \pmod{4}, \\ v_2(q-2), & p = 2, q \equiv 2 \pmod{4}. \end{cases}$$

The logarithm is nonzero: b_p is neither the identity nor a nontrivial torsion element in these principal-unit neighborhoods.

For an ordinary odd integer x ,

$$a^x = \varepsilon_p \exp_p(\ell_p x). \quad (10.2)$$

The tower exponents are all odd, so this is exactly the analytic map that must be continued to the limiting component. At odd p the sign -1 is prescribed by the odd tower branch; it should not be reconstructed from a nonexistent continuous parity function on $\mathbb{Z}_p$.

10.2 The closed formula and its proof

Define the small-argument Lambert series

$$W_p(z) = \sum_{j \geq 1} \frac{(-j)^{j-1}}{j!} z^j. \quad (10.3)$$

Formal Lagrange inversion of $w \mapsto we^w$ gives these coefficients: $[z^j]W(z) = j^{-1}[t^{j-1}]e^{-jt} = (-j)^{j-1}/j!$. The series converges when $v_p(z) > 1/(p-1)$. For the present arguments, this follows directly from

$$v_p\left(\frac{(-j)^{j-1}}{j!} z^j\right) \geq jv_p(z) - \frac{j-1}{p-1} \longrightarrow +\infty.$$

The inverse identity $W_p(z) \exp_p(W_p(z)) = z$ therefore holds in this neighborhood. These facts agree with the standard p -adic Lambert theory in [4].

Theorem 10.1 (Local limit formula). *For every prime $p \mid q$, the component of the stable tower limit is*

$$\boxed{\xi_{q,p} = -\frac{W_p(-\varepsilon_p \ell_p)}{\ell_p}}. \quad (10.4)$$

All series in this expression converge in $\mathbb{Q}_p$.

Proof. The valuations of ℓ_p are at least 1 for odd p and at least 2 for $p = 2$, so $v_p(\ell_p) > 1/(p-1)$. By the exact local distances, T_h converges to $\xi_{q,p}$. Continuity in (10.2) gives

$$\xi_{q,p} = \varepsilon_p \exp_p(\ell_p \xi_{q,p}).$$

Put $w = -\ell_p \xi_{q,p}$. Then $w \exp_p(w) = -\varepsilon_p \ell_p$. The small inverse branch is unique: on its domain the linear term of $w \exp_p(w)$ preserves distances and the higher terms are strictly smaller. Hence $w = W_p(-\varepsilon_p \ell_p)$, proving the formula. Alternatively, substituting the right side of (10.4) into the fixed-point equation and using the strict contraction of $x \mapsto \varepsilon_p \exp_p(\ell_p x)$ proves the same identification. $\square$

Example 10.2 (Why a uniform sign would be wrong). Take $q = 6$, so $a = 5$. The 2-adic component uses $\varepsilon_2 = 1$ and $\ell_2 = \log_2(5)$; it is congruent to 1 modulo 4. The 3-adic component uses $\varepsilon_3 = -1$ and $\ell_3 = \log_3(-5)$; it is congruent to -1 modulo 3. A blind use of the minus-sign formula at both primes would give the wrong 2-adic branch, congruent to -1 modulo 4. Extending the 2-adic logarithm by $\log_2(-1) = 0$ does not repair the mistake: logarithm then forgets exactly the sign that exponentiation of an odd integer retains.

10.3 Reproducible rational-series verification

The function `local_lambert_residue` evaluates finite logarithm and Lambert polynomials using exact rational numbers. It is intended as an independent educational check, not as the fastest algorithm. At a requested p -adic precision K , it sets an auxiliary target of $K + 4$ digits and uses $2(K + 4) + 4$ terms in each series.

Here are sufficient tail estimates. If $s = v_p(b_p - 1)$, a logarithm term of degree j has valuation $js - v_p(j)$. Since $v_p(j) \leq j/2$ for $j \geq 2$, the chosen truncation makes the entire logarithm tail vanish beyond the requested precision. For the fixed-point series obtained by dividing (10.3) by ℓ_p , a term with index j has valuation at least

$$(j-1)s - v_p(j!).$$

For odd p , this is at least $(j-1)/2$; for $p = 2$, where $s \geq 2$, it is at least $j-1$. These bounds justify the chosen truncation. Substituting the truncated logarithm preserves the target precision: the linear coefficient of the fixed-point series is 1, and the higher powers on these principal-unit domains cannot magnify an input error. This can also be checked termwise using $x^k - y^k = (x - y) \sum_{i=0}^{k-1} x^{k-1-i} y^i$ and the same factorial bounds.

The supplied checks compare this rational-series evaluation with the general modular oracle at 60 local parameter triples, including both residue classes of q modulo 4.

11 Conclusions and the remaining boundary of the result

The source problem has a complete resolution in its stated domain. C1 is true, and in fact there is a unique fixed-point residue modulo every q^m . C2 is true, with the stronger explicit formula $T_h \equiv q^2 - q - 1 \pmod{q^2}$ for all $h \geq 2$. C3 is false for every $q = 2n \geq 4$; the error is not merely nonzero but always $-2q^m$ modulo q^{m+1} .

The unified theorem is

$$T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}, \quad h \geq 0, \quad r \geq 1.$$

It yields an exact stabilization threshold, a digit-lifting recurrence, and modular saturation across sufficiently large Knuth arrow ranks. The prime-adic analysis explains why the factor 2

in the leading error is universal and why it never supplies a full additional radix digit when $q \geq 4$.

Several possible continuations are natural, but are not being labelled verified open problems here. One could classify similarly explicit first-unstable-digit laws when the base is congruent to a root of unity other than -1 ; determine sharper sufficient rank bounds than $n + r - 2 \geq m$; or build a finite-precision evaluator for broader classes of chained-arrow expressions. Each would need its own precise source check and proof. Nothing in this note resolves the general analytic extension of tetration, arbitrary-modulus hyperoperation complexity, or the comparison of unrestricted ordinal-indexed fast-growing hierarchies.

A A compact replacement for the OEIS conjectural comments

The following mathematical text is suitable as a proposed correction; it has not been submitted to OEIS by this work.

Let $q = 2n \geq 4$, $T_0 = 1$, and $T_{h+1} = (q - 1)^{T_h}$. For all $h \geq 0$ and $r \geq 1$,

$$T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}.$$

Thus the first two conjectures are true. In fact, $A(m, n)$ is the unique solution of $(q - 1)^x \equiv x \pmod{q^m}$, and $T_h \equiv q^2 - q - 1 \pmod{q^2}$ for $h \geq 2$. The third conjecture is false for every $n > 1$: the correct relation is $A(m + 1, n) \equiv T_m - 2q^m \pmod{q^{m+1}}$. For example, $3 \uparrow\uparrow 2 \equiv 27$ but $3 \uparrow\uparrow 3 \equiv 59 \pmod{64}$.

B Proof-audit checklist

A reader can audit the principal claims without examining the entire software package. The important checks are these.

1. Every T_h is odd, including $T_0 = 1$; every difference used as an exponent in lifting is positive and even.
2. For odd $p \mid q$, $p \nmid q - 2$. The initial difference $T_r - 1$ therefore has valuation zero at p .
3. At 2, the initial valuation is u , not zero; each lift adds $u + v - 1$, not merely v .
4. The exceptional case where q is a power of 2 is treated separately when proving exact radix- q precision.
5. The leading-difference binomial estimate uses $2q^h \mid D_h$, rather than only $q^h \mid D_h$.
6. Exponent reduction is justified by an explicit period identity. The code does not assume that arbitrary modular exponentiation is a function of the exponent modulo the output modulus.
7. A repeated canonical residue $R_m = R_{m+1}$ is not confused with $T_m \equiv T_{m+1} \pmod{q^{m+1}}$.
8. The analytic formula keeps the local sign at 2 separate; the higher-arrow bound excludes the exceptional base $a = 2$.

The detailed companion file `PROOF_AUDIT.md` records the same checks and separates mathematical proof from finite verification.

C Package and reproduction instructions

The archive contains the complete L^AT_EX source, this PDF, two standard-library Python files, generated CSV and JSON data, a source record, and build instructions.

From the extracted directory, run

```
python code/verify.py
python code/tower_digits.py 4 3 --height 2
python code/tower_digits.py 4 3 --height 3
python code/tower_digits.py 10 60 --rank 1000000 --argument 2
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The two finite-height commands print 27 and 59. The third prints the unpadded decimal representative for the example in [section 9](#). The verification script regenerates its report, the residue tables, the local Lambert table, and the counterexample certificate. Elapsed time is machine-dependent; the mathematical outputs and assertion counts are deterministic.

References

- [1] Davis Smith and OEIS contributors, *A324017: Square array of residues of $(2n - 1) \uparrow\uparrow m$ modulo $(2n)^m$* , The On-Line Encyclopedia of Integer Sequences, entry introduced 28 March 2019. <https://oeis.org/A324017>; internal record: <https://oeis.org/A324017/internal>. Retrieved 19 September 2026; the internal entry showed revision #91, Feb 16 2025 and retained the three conjectural comments.
- [2] Marco Ripà, *The congruence speed formula*, arXiv:2208.02622, 2022; journal reference: *Notes on Number Theory and Discrete Mathematics* 27(4), 43–61 (2021). <https://arxiv.org/abs/2208.02622>. Used for the established decimal congruence-speed context, not as a source of a proof of the selected OEIS statements.
- [3] Markus Hittmeir, *A reduction of integer factorization to modular tetration*, arXiv:1707.04919, version 3, 2018; *International Journal of Foundations of Computer Science*, 2020, DOI 10.1142/S0129054120500197. <https://arxiv.org/abs/1707.04919>.
- [4] István Mező, *The p -adic Lambert W function*, arXiv:1801.00657, 2018. <https://arxiv.org/abs/1801.00657>. Used for the established definition and convergence theory of W_p .